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Valorization of lignin into carbon spheres, carbon dots, and graphene-like layers via hydrothermal conversion

Giulia Ischia [a,b,*,1], Akira Yoko [b,c], Wahyudiono [c,e], Luca Fiori [a,f], Tadafumi Adschiri [b,d,*]

[a] Department of Civil, Environmental and Mechanical Engineering, University of Trento, Via Mesiano 77, 38123, Trento, Italy
[b] WPI Advanced Institute for Materials Research (WPI-AIMR), Tohoku University, 2-1-1 Katahira, Aoba-ku, Sendai, Miyagi, 980-8577, Japan
[c] International Center for Synchrotron Radiation Innovation Smart (SRIS), Tohoku University, 468-1 Aramaki-Aza-Aoba, Aoba-ku, Sendai, 980-8572, Japan
[d] New Industry Creation Hatchery Center, Tohoku University, 6-6-10 Aoba, Aramaki, Aoba-ku, Sendai, 980-8579, Japan
[e] Department of Chemical Engineering, Institut Teknologi Sepuluh Nopember, Kampus ITS Sukolilo, Surabaya, 60111, Indonesia
[f] Center Agriculture Food Environment (C3A), University of Trento, San Michele all'Adige, 38010, Trento, Italy

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ABSTRACT

Lignin is a renewable, aromatic-rich byproduct of biomass processing, and holds a significant potential as a precursor for sustainable carbon materials. This study presents a proof of concept for its conversion into diverse carbon-derived products via a two-stage hydrothermal process. In the first hydrothermal stage (250 °C, 60 min), lignin undergoes fragmentation and partial depolymerization, yielding a liquid phase rich in soluble aromatic fragments and a minor solid residue (5.2 %). In the second hydrothermal stage (300–350 °C, 15–60 min), aromatic precursors undergo conversion into carbon spheres (180–220 nm), carbon dots (<10 nm), and graphene-like layers. The carbon spheres form mainly via self-assembly and condensation, with a carbon content of 72.5 % and a mass yield of 42.3 %. Carbon dots, which confer photoluminescence, partially coalesce into larger structures, contributing to carbon sphere formation as a secondary pathway. Simultaneously, aromatic fragments undergo rebonding and planar fusion, reorganizing into graphene-like layers with localized crystalline domains. All carbon-derived products form within the first 15 min of the second stage, with longer times promoting minor structural rearrangements. Overall, the proposed chemical-free hydrothermal route represents a green strategy for valorizing lignin into functional carbon materials, serving as a basis for further investigation into their formation mechanisms and applications.

1. Introduction

Lignin is the largest renewable source of aromatics on Earth, accounting for 15–40 % of plant biomass [1]. It is a highly irregular, oxygenated biopolymer of polyphenolic units formed through the polymerization of monomeric precursors in plant cell walls, such as *p*-hydroxyphenyl, guaiacyl, and syringyl units [2]. It consists a network of carbon-carbon and ether bonds, which support mechanical strength and confer rigidity to plant cell walls and resistance to microbial attack [3]. While lignin can be directly extracted from plants using solvent, acidic, or alkaline methods [4,5], it is also obtained as a byproduct of the pulping and bioethanol industry — often referred to as "technical lignin" [6]. Currently, waste lignin accounts for approximately 70 million tons annually [7,8], most of which is either discarded or burned onsite to power the plants [9]. However, lignin has potential far beyond mere combustion; its highly aromatic molecular structure and carbon content make it an excellent candidate for producing carbon materials for various applications. These last include for example energy storage [7,10–12], environmental remediation [13], catalysis [14], biomedical uses [15,16], or wearable electronics [17]. In this regard, recent research has been conducted on converting lignin into nanocarbons, driven also by the growing interest in bio-based materials. For example, lignin was successfully converted into spherical nanoparticles [18], graphene nanolayers [19], carbon dots [20], carbon fibers [21],

* Corresponding authors at: WPI Advanced Institute for Materials Research (WPI-AIMR), Tohoku University, 2-1-1 Katahira, Aoba-ku, Sendai, 980-8577, Miyagi, Japan.
E-mail addresses: giulia.ischia@mpikg.mpg.de (G. Ischia), tadafumi.ajiri.b1@tohoku.ac.jp (T. Adschiri).
[1] Present address: Department of Colloid Chemistry, Max Planck Institute of Colloids and Interfaces, Am Mühlenberg 1, 14476, Potsdam, Germany.

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nitrogen-doped hierarchical carbons [22], and hydrogels [23,24]. The potential also led to a few noteworthy commercialization cases, like the Lignode® from Stora Enso as anode replacement [25].

Significant challenges remain in converting lignin into submicron-sized carbon materials. Indeed, the variable macromolecular structure of lignin and its poor solubility in water hinder scalability and performance [26]. Meanwhile, current methods often rely on techniques that are arguably not environmentally friendly, being chemical-intensive, corrosive, and requiring high temperatures [7]. For example, lignin carbon fibers are commonly produced by spinning precursors such as polyacrylonitrile or pitch into fibers, followed by thermal stabilization, making the process highly energy-intensive [7]. The synthesis of graphene-like lignin, as well of other biomasses, often involves processes such as chemical exfoliation, chemical vapor deposition, epitaxial growth, or pyrolysis [27,28]. Meanwhile, lignin spherical particles often rely on solvent exchange or self-assembly techniques [29,30], which, although less intensive, require the use of organic solvents, and lack of precise control and scalability. Indeed, as a counter-effect, greener alternatives often result in products with lower quality or stability [31,32]. Therefore, there is a need to find a balance between sustainability, performance, and scalability — a challenge that can open opportunities for new exciting research. In this context, hydrothermal conversion processes stand out as a green biomass valorization technique. In the last decade, they have gained significant attention for their versatility in biomass processing and the synthesis of carbon-materials [33,34], and even metal oxide nanoparticles and metal-organic frameworks [35,36]. These processes exploit the unique properties of hot pressurized water (180–373 °C, 10–220 bar in subcritical), which acts similarly to an organic solvent, favoring the breakdown and conversion of biomass while promoting the synthesis of carbon-based products [37,38]. Depending on the operating conditions, they are commonly classified as hydrothermal carbonization (HTC, 180–250 °C), hydrothermal liquefaction (HTL, 280–373 °C), and hydrothermal gasification under supercritical conditions (>373 °C) [39]. Table 1 summarizes the common applications of hydrothermal processes for carbon materials from biomass, with applications spanning from energy storage and catalysis to drug delivery and environmental remediation. Most studies have focused on the HTC (in batch reactors) of carbohydrate-rich substrates, which are well known for their ability to form carbon microspheres via the back-polymerization of furan intermediates [40–42]. Process variants, such as the decoupling of temperature and pressure in semi-continuous apparatus, have also been investigated for the rapid conversion of cellulose into microspheres [43]. However, the microspheres obtained from direct HTC are typically not used as final products, but rather serve as intermediates requiring subsequent treatments to tailor porosity, graphitic order, and stability. For example, HTC combined with pyrolysis (500–700 °C) produces carbons with enhanced graphitization and porosity [40,44]. Chemical activation (with KOH) is often used to produce highly porous carbons with improved conductivity for electrochemical devices [45,46], while physical activation with CO_2 for porous spheres performed less well but with lower impacts [40,47]. Moreover, templating approaches (for example, using SiO_2 or surfactants) were used to produce hollow or mesoporous spheres with high surface areas and controlled pores [40]. In parallel, HTC has been widely employed for the synthesis of carbon dots from diverse biomasses, yielding <10 nm particles with excitation-dependent photoluminescence [48,49]. By contrast, the application of hydrothermal processes to lignin carbon materials has been far less explored [9,42]. For example, modified HTC with H_2O_2 was used to produce graphene carbon dots with tunable optical properties [50]. Recent studies have

Table 1
Summary of current representative studies on the hydrothermal conversion of biomass and lignin in carbon spheres, carbon dots, and graphene-like layers. Process types, operating conditions, substrates, applications, and main drawbacks are reported.

Process	Operating conditions	Substrate	Properties, advantages	Disadvantages	Ref.
Carbon spheres					
Hydrothermal	180–250 °C, 1–24 h	Carbohydrates	Smooth morphology; simple, chemical-free process; precursors for adsorbents, catalysts, and electrodes.	Low porosity and limited surface area unless further treated	[54–56]
Decoupled hydrothermal	20 MPa; no hold time	Cellulose	Submicron in very short times; semi-continuous.	Specialized high-pressure equipment; limited demonstration at scale.	[43]
Hydrothermal → pyrolysis	HTC 180–250 °C 2–24 h → pyrolysis 500–700 °C 2–7 h)	Carbohydrates	Enhanced graphitization and porosity; suitable for catalytic supports and ethanol conversion.	Energy intensive due to high-temperature second step	[40,44]
Hydrothermal → templating	180–200 °C, 12–24 h + SiO_2, surfactants	Carbohydrates, phenolics	Hollow or mesoporous spheres; high surface area and controlled pore structure, useful for drug delivery, catalysis, and energy storage.	Requires template removal (HF/HCl or calcination), limiting scalability and greenness	[40,57,58]
Hydrothermal → chemical activation	180–290 °C 7–12 h → 700–800 °C 1–4 h with KOH	Carbohydrates, enzymatic hydrolysis lignin	Highly porous activated carbon; high surface area and conductivity, suitable for supercapacitors and electrochemical devices.	Use of corrosive chemicals (KOH); multi-step process; safety and waste management issues.	[45,46]
Hydrothermal → physical activation	180–220 °C 1–6 h CO_2 at 800 °C 0–2 h	Carbohydrates, grass	Porous; milder activation compared to KOH, improved sustainability; applications in adsorption and electrodes.	Lower surface area than chemical activation; requires continuous gas supply.	[40,47]
Carbon dots					
Hydrothermal	160–220 °C, 2–12 h, sometimes with dopants	Various biomasses	<10 nm particles; excitation-dependent photoluminescence; suitable for imaging, sensing, photocatalysis.	Broad size distribution; often low quantum yield without surface passivation.	[16,48,49,59]
Hydrothermal in H_2O_2	180 °C, 10–60 min	Lignin	Blue emission, high hydrophilicity and biocompataibility	Oxidants; possible surface over-oxidation; low carbon efficiency.	[50]
Hydrothermal (supercritical)	450 °C 24.8 MPa, 2 min	Glucose	Formation of carbon quantum dots, yields ~4× higher than subcritical runs; sharp PL peak at 446 nm (excitation 360 nm).	Requires high-pressure equipment; elevated technological complexity	[60]
Graphene-like layers					
Modified hydrothermal	160 °C 15 min, in NaOH	Pulping black liquor	Hetero-junctions for supercapacitor. Micron-sized, green, scalable.	Few studies, requires alkaline medium.	[19,51]
	90 °C 8 h with NaOH + HCl	Lignin with graphene oxide	Lignin acts as reducing agent for GO, yielding reduced graphene oxide composites.	External GO precursor; moderate conductivity.	[53,61]
Modified hydrothermal → pyrolysis	HTC 180 °C, 48 h → Pyrolysis 700 °C, 8 h, with Fe_3O_4	Soda papermarking waste liquor + Fe_3O_4 particles	Partial graphitization; potential in catalysis and composites.	Multi-step; high energy demand; relies on added catalysts.	[52,53]

also demonstrated that lignin can be converted into graphene-like layers and graphene carbon dots under a modified hydrothermal environment with KOH/NaOH, highlighting the ability of hydrothermal of induce covalent cross-linking and aromatic condensation [19,51]. Also, modified hydrothermal conditions with additional agents (e.g., alkali, Fe_3O_4) and a post-pyrolytic stage were used to produce graphene-like carbons [52,53]. Overall, these strategies, although effective, require the integration of different technologies (increasing complexity, and studies targeting lignin remain scarce and rely on chemical additives or multi-step processes.

Therefore, aiming to exploit the unique potential of lignin via new green routes, we present a novel method for converting lignin into a variety of carbon-derived products relying solely on hot-pressurized water in subcritical conditions. The conversion scheme involves a two-stage hydrothermal process. The first stage (250 °C, 60 min) allows the defragmentation and partial depolymerization of lignin into aromatic precursors, while the second stage (300–350 °C, 15–60 min) is for the synthesis of carbon spheres, carbon dots, and graphene-like layers. Dividing the process into two stages allows for better control of the reaction. By analyzing the solid and liquid phases over the reaction time, we demonstrate the effectiveness of the process and gain insights into the underlying formation mechanisms. This study thus provides a proof of concept for a novel two-stage hydrothermal route to valorize lignin into carbon materials, opening the way for further optimization and potential applications.

2. Material and methods

2.1. Hydrothermal runs

The conversion scheme involves a two-stage hydrothermal process. The first stage (250 °C, 60 min) consists of the defragmentation and partial depolymerization of lignin, yielding a liquid phase enriched in aromatic fragments. This liquid phase (free of residual solid) is then subjected to the second stage (300–350 °C, 15–60 min), during which three product fractions form concurrently, defined operationally as follows: (i) carbon spheres (CS), the solids retained on 0.025 μm membranes (used to compute CS yield); (ii) carbon dots (C-dots), <10 nm present in the permeate/supernatant; and (iii) graphene-like layers, obtained by freeze-drying the retentate after removal of CS and insolubles with a 1 μm cut-off.

Each stage consists of runs performed in batch-type reactors comprising a tube body and cap with an internal volume of 5.0 mL (Hastelloy C-276, AKICO Co. Ltd., Japan) [62]. The procedure consists of: loading the slurry into the reactor, heating it in a mechanically shaken electrical furnace to the set-point temperature (heating time: 2–3 min), rapidly quenching the reactor to suppress repolymerization in a cold-water bath after the residence time, and releasing the gas [63]. For the first stage, every reactor was loaded with 0.12 g of lignin (dealkali, TCI) and 4 mL of distilled water corresponding to a mass biomass to water ratio (B/W) of 0.03. The values was selected after preliminary runs at higher B/W (0.1), typically adopted in hydrothermal conditions [63–65]. The higher dilution suppressed coagulation/aggregation of lignin particulates and favored dissolution into the liquid phase, in line with reported dilution effects in hydrothermal media [66]. Principal runs were performed at 250 °C for 60 min, a condition known from the literature to initiate lignin fragmentation while avoiding carbonization [67]. Extra runs were performed at 300 °C 5 min, 350 °C 24 h, and 375 °C 60 min were included to span the severity range where lignin undergoes hydrothermal conversion (>250 °C) [63]. The solid and liquid phases were separated using a qualitative filter (Whatman, 1 μm) and collected. The solid phase was weighted and used to compute the solid yield (SY) as the mass ratio of the residual solid over the initial lignin, while the liquid phase was used for the second stage. For the second stage, the liquid phase from the first stage was subjected to a second hydrothermal run. Each reactor was loaded with 2 mL of liquid phase diluted at a volume ratio of 1:1 with distilled water. Temperatures were set at 300, 320, and 350 °C, and residence times at 15, 30, and 60 min. Temperatures were chosen as known to promote the covalent cross-linking during hydrothermal conversion from the literature, while residence times are typical of hydrothermal processing of biomass precursors [19,68]. CS were recovered via a 0.025 μm hydrophilic membrane filter (Millipore), dried overnight, weighed, and their SY computed. For C-dots and graphene-like layers, the slurry consisting of solid and dissolved compounds was filtered at 1 μm, washed with deionized water, diluted, sonicated for 60 min, and centrifugated (1000 rpm for 30 min) to remove CS and insoluble fractions [19]. The resulting suspension was freeze-dried, and the solid phase recovered. To further investigate the formation of graphene-like layers, additional runs were performed on the liquid phase from the first stage, with the addition of pure toluene (volume ratio 1:1) at 250, 320, and 350 °C for 60 min. Each condition was performed in at least five independent runs ($n \geq 5$), and data are presented as the mean with the corresponding standard deviation. For a small set of pre-specified contrasts (e.g., 300 vs 350 °C at fixed time; 15 vs 60 min at fixed temperature), differences were evaluated using Welch's two-sample *t*-test (two-sided, $\alpha = 0.05$); when multiple contrasts were performed within the same panel, Holm adjustment was applied.

2.2. Solid phase analyses

Composition analysis was conducted for C, H, N, and O; C was measured by combustion (2200 W) and infrared detection method (CS844, LECO Corp., Saint Joseph, MI, United States). The samples were loaded on ceramics crucible (LECO 528-018HP) with 1.5 g of W particles and 0.5 g of Sn particles in 0.5 g of Ni capsule. The calibration was conducted with the standard material (JSS208-4, C:1.017 %). H was measured by Ar transfer melting (3000 W, approximately 2000 °C) — thermal conductivity method (EMGA-821, HORIBA, Ltd., Kyoto, Japan). The samples were loaded on graphite crucibles (HORIBA 3200043543) with 1 g of Sn particles. The Ar carrier gas was introduced at 420 mL/min, and the H_2 gas (>99.99999 %) was used for the calibration standard. N and O were measured with He transfer melting (4800 W, approximately 2165 °C), thermal conductivity method and infrared spectroscopy method, respectively for N and O with the analyzer (ON836, LECO Corp., Saint Joseph, MI, United States). The samples were loaded on graphite crucibles (LECO 782-720) with 1 g of carbon and 0.2 g of Sn pellets. The standard sample, LECO YY-001-103 (Lot No. 103-42) that includes 179 ± 2 ppm of O and 60 ± 2 ppm of N was used. The flow rate of the He carrier gas was set at 460 mL/min. Thermogravimetric analyses (TGA) were performed using an STA 2500 Regulus Netzsch system coupled with a gas chromatography (GC) system (Agilent Technologies 7890B) and mass spectroscopy (JMS-Q1500GC, JEOL). The temperature program included heating from 50 to 110 °C (held for 30 min), then from 110 to 900 °C at a rate of 10 K/min under an inert argon (>99.9999 vol%) atmosphere at 50 mL/min, followed by a transition to air from 900 to 950 °C for ash determination. The mass loss observed between 900 and 950 °C was attributed to fixed oxidizable carbon (FC) [67]. Each sample was analyzed in at least three technical replicates; results are reported as mean with the corresponding standard deviation. For targeted pairwise comparisons of composition (e.g., C content, H/C, O/C) or FC between two conditions, differences were evaluated using Welch's two-sample *t*-test (two-sided, $\alpha = 0.05$) on replicate-level data; when multiple contrasts were performed within the same figure or table, Holm adjustment was applied.

Fourier-transform infrared (FT-IR) spectra were obtained using a FT/IR-4200 spectrometer (JASCO Corp.) with samples prepared via the KBr pellet method. The morphology was examined using field-emission scanning electron microscopy (FE-SEM; JEOL JSM-7800F) at an electron beam energy of 5–15 keV and transmission electron microscopy (TEM) with a Hitachi H-7650 instrument at 100 kV and a JEOL JEM-ARM200F instrument at 200 kV. Amorphous versus crystalline

structures properties were characterized using confocal Raman microscopy (HORIBA LabRAM HR-800, Japan) with a 532 nm excitation wavelength. Crystal structure analysis was conducted using X-ray powder diffraction (XRD; Rigaku SmartLab 9MTP, Japan) with Cu Kα radiation (λ = 1.5418 Å). XRD patterns were recorded across the 2θ range of 10°–80° at a scanning speed of 1.5°/min.

2.3. Liquid phase analyses

The composition of the liquid phases was analyzed using a Shimadzu High-Performance Liquid Chromatography (HPLC) system equipped with a Kinetex F5 2.6 μm column (Phenomenex Inc., CA, USA). The mobile phase consisted of methanol and water in a 10:90 volume ratio, operated in isocratic mode with a flow rate of 1.0 mL/min. Detection was performed at 280 nm using a UV–Vis detector (UV-2070 Plus, Jasco, Tokyo, Japan), with the column maintained at 40 °C in an oven (CO-2065 Plus, Jasco, Tokyo, Japan). Samples were introduced via an autosampler (AS-2075 Plus, Jasco, Tokyo, Japan), and data acquisition was managed with ChromNav 2.04 software (Jasco, Tokyo, Japan). Total organic carbon (TOC) of the liquid phases from the first and the second stage was measured by combustion and CO_2 detection with Non-Dispersive Infrared Absorption method (TOC-L, Shimadzu Corp., Japan). The combustion temperature was set at 680 °C. Photoluminescence (PL) of liquid samples containing C-dots was investigated via Spectrophotometer Hitachi High-Technologies, F-2700, at excitations wavelengths between 220 and 600 nm. Each sample was analyzed at least three times and mean values with corresponding standard deviation are reported.

Carbon mass balances were made using the parameter carbon migration efficiency (CE), representing the fraction of carbon migrating from lignin to the solid and liquid phases was referred to as, which was computed as:

$$CE_{liquid} = \frac{TOC}{B/W} \bullet 1000 \quad (1)$$

$$CE_{solid} = \frac{C_i \bullet SY}{C_{bio}} \quad (2)$$

where C_i the C content in the i-solid phase, C_{bio} the C content in the initial biomass. SY of the freeze-dried suspension was computed as the ratio between CE_{liquid} and C_{bio}. We report uncertainties calculated via first-order error propagation, using the measured standard deviations of the input terms.

3. Results and discussion

3.1. Lignin conversion to aromatic precursors

The first hydrothermal stage (at 250 °C 60 min) was performed to depolymerize and defragment lignin into the aqueous phase. The low amount of recovered solid (SY of 5.2 %, Table 1) confirms its effectiveness. The resulting slurry is dark-colored (**Fig. S1**). Such a low SY deviates from typical findings in the hydrothermal conversion of lignin at mild temperatures (180–250 °C, the typical carbonization regime), which generally requires temperatures above 280–300 °C to achieve significant conversion [67]. In this case, the highly diluted environment (with a B/W of 0.03) appears to be a critical factor. To exclude other parameters, we performed an additional run at a B/W of 0.1, which indeed resulted in a higher SY of 51.5 %, consistent with literature values [64,67]. This phenomenon arises from the high dilution, which creates an almost homogeneous environment where powdered lignin is highly dispersed in water, increasing surface exposure. This promotes solid-liquid reactions, such as hydrolysis and water solvation effects, while suppressing lignin coagulation—ultimately favoring its dissolution [31]. Then, another factor is rapid cooling at the end of the process, which prevents repolymerization, inhibits the self-assembly of lignin fragments, and slows oxygen-loss reactions such as dehydration [69]. The residual solid phase recovered from filtration resembles the original lignin (Table 1 and Fig. S2). Its C content slightly increases from 51.4 to 54.5 % ($p < 0.01$, two-tailed *t*-test), and the DTG peak passes from 335 °C for lignin to 377 °C for the solid residual, indicating that no carbonization occurs, which would need indeed higher temperatures (>280–300 °C) [63,67]. Also, the residual solid contains no ash, indicating that the inorganic fraction present in the original lignin (mainly consisting of Na and Si [70]) has migrated entirely into the liquid phase during the process.

TOC data (Table S1) on the liquid phase confirm the near-complete dissolution of lignin. Similarly, carbon migration efficiency data (CE, computed using Eq. 1) indicate that 92.1 % of C migrates from lignin to the liquid phase (Table S1). The HPLC spectrum (Fig. 1) reveals that the liquid phase is dominated by undetectable fragments, along with smaller amounts of compounds such as guaiacol, 4-ethylguaiacol, and syringaldehyde, which are typical products of lignin depolymerization [71]. The undetected fragments consist of high-molecular-weight compounds soluble that remain relatively large and structurally complex, not fully depolymerized. Due to their size, they elute quickly in HPLC because they interact minimally with the stationary phase, passing through without significant separation, remaining undetected. Their presence suggests that the aromatic backbone of lignin remains preserved throughout the conversion and undergoes reactions such as the cleavage of its carbon side chains, demethylation, and partial breakdown of β-O-4 bonds. Together with the detected monomers, these fragments form reactive, soluble precursors with amphiphilic character and exposed functionalities, providing a tunable feedstock for the self-assembly and condensation processes of the second stage.

Therefore, the first stage favors the dissolution of lignin without significant re-organization of the main system, leading to aromatic. Compared to traditional techniques, we benefit from a chemical-free green reaction environment and precursors free of organic solvent. Although it can have drawbacks like increased energy consumption and reduced productivity, high dilution appears crucial for maximizing the number of precursors and to control the subsequent stage.

3.2. Carbon spheres: morphology, yields, and formation

CS were recovered as the solids retained on 0.025 μm hydrophilic membranes after the second hydrothermal stage (300–350 °C, 15–60 min). CS exhibit a regular spherical morphology with diameters of 180–220 nm and a smooth surface, as shown in the SEM and TEM images (Fig. 2a-c). The spherical shape is maintained across operating conditions, with no signs of agglomeration between adjacent CS. SYs range from 40.0–42.3 % at 320 °C to 31.5–34.0 % at 350 °C, with slight variations depending on residence time (Table 2). The decrease in SY at higher temperatures is primarily attributed to increased gas production, as higher severities favor decarboxylation and decarbonylation reactions. The presence of CS at short residence times (e.g., 15 min) suggests that primary conversion occurs in the early stages of the reaction. Therefore, the main structural transformations likely begin during the

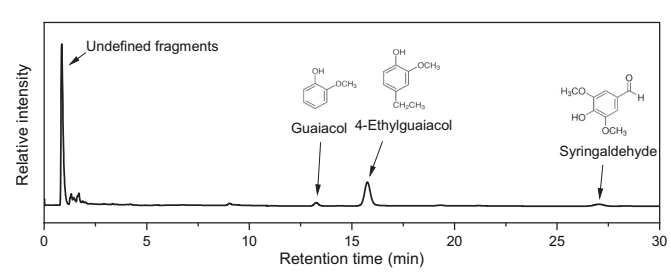

Fig. 1. HPLC spectrum of the liquid phase recovered after the first hydrothermal stage (250 °C, 60 min).

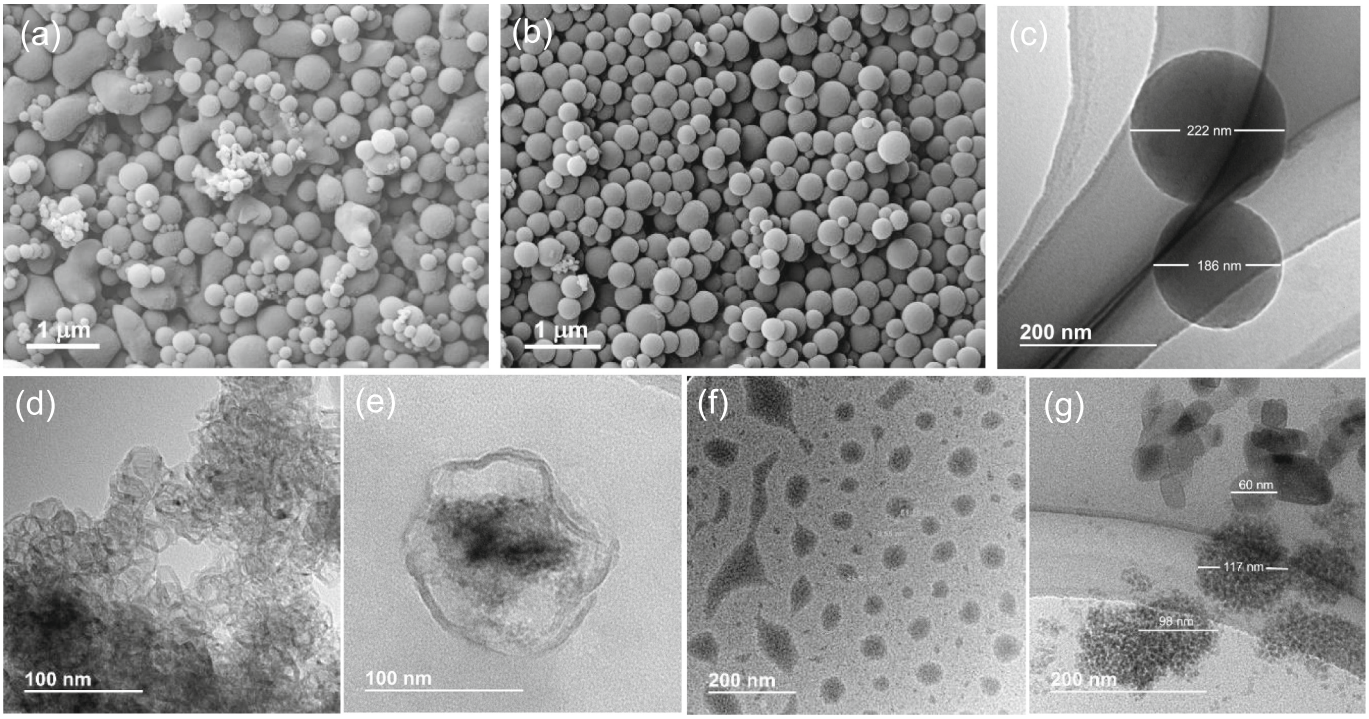

Fig. 2. SEM and TEM images of CS and C-dots aggregates obtained under different hydrothermal conditions: (a–c) CS formed at 320 °C for 15 min, 320 °C for 60 min, and 350 °C for 15 min, respectively; (d–e) close-up views of self-assembly features observed at 300 °C for 60 min; (f–g) aggregates of C-dots observed at 320 °C for 15 min.

Table 2
Properties of lignin, residual solid phase from the first stage, and CS from the second stage produced at different temperature and reaction times. Ash content: 6.9 % for lignin, 0 % for other phases. Confidence intervals represent standard deviations computed from error propagation (in details for SYs, for elemental <0.5 %, FC < 1.0 %, H/C and O/C < 0.1). Data are referred on a mass base.

Sample	Temp. (°C)	Time (min)	SY (%)	Elemental (%)				FC (%)	H/C (–)	O/C (–)
				C	H	N	O			
Lignin	–	–	–	51.4	3.6	0.2	38.1	41.6	1.0	0.6
Residual solid	250	60	5.2 ± 0.2	54.5	3.2	0.2	42.1	54.8	0.8	0.7
CS	320	15	40.6 ± 0.8	67.8	3.5	0.4	28.3	56.8	0.7	0.4
		30	40.0 ± 0.6	69.3	3.1	0.4	27.2	56.2	0.6	0.3
		60	42.3 ± 0.5	71.0	3.1	0.4	25.5	57.4	0.6	0.3
	350	15	33.9 ± 0.4	68.2	3.2	0.3	28.3	57.9	0.6	0.4
		30	34.0 ± 0.7	72.2	3.0	0.3	24.5	60.5	0.6	0.3
		60	31.5 ± 0.8	72.5	3.0	0.5	24.0	66.3	0.6	0.3

initial conversion phase and/or the heating period. This observation agrees with a liquid-to-solid formation mechanism, in which the homogeneous reaction environment (consisting of lignin precursors in the aqueous phase) facilitates rapid interactions between dissolved lignin fragments. The SY values of 31.5–42.3 % are within the range of HTC carbons from carbohydrates (30–50 %) [54,56]. In contrast, solvent-exchange routes for lignin nanoparticles typically recover only a limited fraction of the initial lignin as particulate solids (often 10–25 % [18], since a large share remains dissolved or is lost during solvent handling.

Data show that both a double-stage process and proper tuning of operating conditions are necessary to obtain regular CS. For example, well-defined spherical CS are produced at 320 °C for 60 min (Fig. 2b) and 350 °C for 15–60 min (Figs. 2c and S3). Skipping the first stage leads to significant morphological deviations, resulting in irregular structures (Fig. S2), which highlights its role in producing CS precursors, minimizing interference from undefined residues, and providing a more controlled input for the second stage. Higher temperatures enhance fragment mobility and accelerate condensation reactions, resulting in more regular CS under conditions such as 320 °C for 60 min and 350 °C for 15–60 min. At low temperatures, like 300 °C, CS exhibit smaller diameters and irregular shapes and form loose agglomerates (Fig. S3), likely due to insufficient energy to drive effective lignin fragment assembly. Conversely, more extreme conditions, such as 350 °C for 24 h or supercritical conditions at 375 °C for 60 min, result in the shrinkage or collapse of CS (Fig. S3).

Regarding formation mechanisms, we can identify several phenomena. TEM images (Fig. 2d–e) show the presence of external shells containing organized fragments, probably indicating self-assembly. Indeed, such behavior agrees with the natural tendency of lignin, which has amphiphilic properties, to form spherical particles [30]. In aqueous environments, the hydrophobic components of lignin aggregate to form the particle core – driven by intermolecular interactions like van der Waals forces and π–π stacking – while the hydrophilic phenolic and aliphatic hydroxyl groups orient outward, forming the particle shell [72]. In addition, TEM images reveal the presence of C-dots (diameters <10 nm), which also form aggregates at various process stages (Fig. 2f–g and Fig. S4). Their appearance at short residence times, small size, and tendency to form agglomerates with larger CS suggest that C-dots may act as nucleation sites or intermediates during CS formation, suggesting

that a secondary formation mechanism likely coexist and/or partially overlaps with self-assembly [19]. In this regard, condensation reactions (occurring in the second stage) may initiate within self-assembled domains (formed during the heating phase) or on C-dots clusters, progressively stabilizing the structure through covalent cross-linking. Indeed, conditions of 300–350 °C (85–133 bar) are severe enough to induce the formation of C—C and C—O—C bonds [73], leading to an improved thermochemical stability (as confirmed in Section 3.3). This thermal stability also allows the CS to withstand the severe hydrothermal environment. Indeed, hot-pressurized water at 320–350 °C exhibits improved solvent-like properties, including a dielectric constant similar to that of organic solvents [39], which would degrade a spherical morphology solely based on self-assembly [18].

3.3. Properties of carbon spheres

As shown in Table 2, as severity increases, the C content in CS increase, with a maximum of 72.5 %, vs lignin at 51.4 %, accompanied with a decrement of O (up to 24.0 %) and a slight decrease in H. These changes result in lower H/C and O/C ratios (from 1.0 in lignin to 0.6–0.7 in CS), indicating the loss of oxygenated groups and supporting the formation of condensed aromatic domains. The marked drop in O/C suggests deoxygenation and enhanced conjugation, consistent with aromatization and improved structural stability. In contrast, variations among the CS are minimal, suggesting that most structural transformations occur during the initial stages of the process, with only minor adjustments taking place thereafter [33]. C of CS is significantly higher than that of common self-assembly methods, such as solvent-shift techniques (Table 3, typically of 50.1–63.3 % [18,74]), supporting the abovementioned hypothesis that in addition to self-assembly, condensation and partial aromatization occur during CS formation. Although the C content of the CS remains nearly the same at both 320 and 350 °C, the significantly higher SYs at 320 °C indicate that a greater portion of the original lignin mass is retained in the solid phase. This suggests that more C from the liquid phase is effectively incorporated into the CS at the lower temperature. This interpretation is supported by the higher parameter CE (calculated using Eq. 2), which reaches 58.4 % at 320 °C compared to 44.4 % at 350 °C after 60 min (Table S1). Thus, higher temperatures favor gas production (originating directly from the liquid phase) rather than enhancing CS formation, in agreement with the favored thermodynamics toward CO_2 and CO formation at elevated temperatures. Moreover, CS do not contain ash, indicating that impurities such as Na and Si do not participate in the repolymerization stage, which is advantageous for a purer final product.

DTG curves (Fig. 3a) confirm the improvement in the thermal stability of the solid products throughout the conversion. The main DTG peak shifts from 335 °C in lignin to 415–500 °C in the CS. At 320 °C, CS show no significant differences across residence times and exhibit a broad, low-intensity decomposition at 500–600 °C. For CS at 350 °C, the thermal stability increases further, with the 15 min sample displaying a DTG peak similar to that at 320 °C but with lower intensity, while the 60 min sample shows a broad decomposition hump at 450–550 °C. This trend, along with the increase in FC from 56.8 to 66.3 %, further supports the hypothesis of enhanced condensation followed by progressive aromatization with increasing severity. Moreover, CS show significantly improved thermal stability compared to those deriving from self-assembly (typically with DTG peaks below 400 °C [29,75]), making them more suitable for various applications such as energy storage, catalysis or electronics [4].

Fig. 3b shows FTIR spectra of lignin and CS at different conditions. Samples exhibit the following vibration peaks: band 3500–3000 cm^{-1} of hydroxyl groups, 2900 cm^{-1} C—H stretching of methyl or methylene groups, 1690 cm^{-1} C═O stretching, 1600 cm^{-1} aromatic skeleton C═C vibration, 1514 cm^{-1} C—C stretching aromatic skeleton, 1460 and 1425 cm^{-1} C—H of methoxy groups, 1332 cm^{-1} syringyl units vibration, C—O at 1265 and 1220 cm^{-1}, 1118 cm^{-1} aromatic C—H deformation in

Table 3
Comparison of lignin-derived CS, C-dots, and graphene-like layers from different common valorization routes. Representative properties with main advantages and drawbacks are listed. Operating conditions are detailed in Table 1.

Process	Properties	Applications, advantages/ disadvantages	Ref.
Carbon spheres			
Hydrothermal → activation	Size 2–6 μm; BET up to 1278 m^2/g	Electrochemical applications + High porosity and conductivity. − Requires corrosive KOH and multistep.	[46]
Hydrothermal → pyrolysis	N-doped CS; ultramicroporous (<0.7 nm); large interlayer spacing	Na-ion battery anodes. + High reversible capacity and stability. − Energy intensive (1100 °C).	[22]
Thermostabilization → carbonization	Size 50 μm; SSA 1255–1939 m^2/g; N 1.1–1.6 at.%	Supercapacitors. + High capacitance and cycling stability. − Large particle size, less suitable for nanoscale uses.	[82]
Solvent-shifting	Size 100–800 nm; C 50–63 %; yield 10–30 %	Drug delivery, UV-blocking, coatings. + Tunable size, high yields. − Lower thermal stability than HTC carbons.	[18,74]
Solvent-shifting → carbonization	Size 250–420 nm	Electrodes, hydrophobic coatings. + Good electrochemical performance. − Two-step, energy demanding.	[29]
Aerosol processing	Size 200–1000 nm; smooth; size tunable.	Pickering emulsions, coatings. + Continuous and scalable. − Requires dedicated atomizer/reactor.	[86]
Acetylation + hydrothermal	N-doped, OH-rich surface; photocatalytic response under visible light	Photocatalysis. + Enhanced activity, self-propelling particles. − Needs acetylation/ solvents.	[87]
Deep eutectic solvent extraction → self-assembly	UV-shielding efficacy and antioxidant functionalities.	Additive for multifunctional nanocomposites + Enables DES-based lignin use. − Requires pretreatment step.	[88]
Two stage hydrothermal	Size 180–220 nm; C up to 72.5 %; high thermal stability	+ Water-only, chemical-free; nanoscale control. − Moderate porosity, defect-rich.	This study
Carbon dots			
Modified hydrothermal (H_2O_2)	2–10 nm; blue PL; hydrophilic; 82.6 % C	Bioimaging/ biosensing; + Simple; strong PL & biocompatible; − Low carbon efficiency, broad size.	[50]
Assisted hydrothermal (amine)	Graphene quantum dots; strong PL; 4× higher fluorescence	+ High photoluminescence − Lab-scale synthesis;	[89]

(continued on next page)

Table 3 (*continued*)

Process	Properties	Applications, advantages/ disadvantages	Ref.
		stability not fully assessed.	
Solvothermal (acids)	~8–9 nm; red/NIR emission (650–710 nm); crystalline	+ Thermal energy storage. − Requires acids; removal of residues.	[90]
Pyrolysis → extraction	Blue PL (~365 nm), QY up to 23 %; O-rich surface	+ Simple, solvent-free, reusable solvent. − Moderate QY, requires high T.	[91]
Antisolvent → carbonizatin	Broad UV–vis absorption; QY ~13 %; emission up to 560 nm	Bioimaging, biosensing. + Stable across pH/ NaCl, low toxicity. − Multi-step, strong acids.	[92]
Two stage hydrothermal	<10 nm; excitation-dependent PL; red emission (620–700 nm)	+ Unique lignin-derived emission. − Low carbon yield; evolve into larger structures at high T.	This study
Graphene-like layers			
Modified hydrothermal (black liquor)	Micron-sized graphene sheets (1–3 layers, 0.36–2.5 μm); transparent conductive films (2.1 kΩ sq^{-1}, 85 % transmittance)	+ Green, scalable, flexible electrodes possible. − Requires alkaline medium; moderate conductivity.	[19,51]
Thermochemical (HTC + pyrolysis + Fe_3O_4)	Lamellar GCS/Fe_3O_4 nanocomposite; donut-shaped Fe_3O_4 embedded in graphene-like sheets; ultrahigh initial Li-ion battery capacity (3829 mAh g^{-1})	+ Very high reversible capacity and long-term cycling stability; waste valorization. − High energy input (700 °C); requires catalyst.	[52]
Modified hydrothermal → pyrolysis	Partial graphitization; Fe_3O_4 nanoparticles in lignin-derived carbon sheets	+ Catalytic and composite potential. − Multi-step, high energy, relies on added catalysts.	[52,53]
Two stage hydrothermal	Amorphous GO-type sheets; C/O ≈ 2.5; G/D ≈ 1.8	+ First additive-free hydrothermal route from lignin. − Defect-rich, probable moderate conductivity.	This study

syringyl unit, 1030 cm^{-1} stretching of primary alcohol [76]. Variations in spectra under different conditions confirm that increasing severity causes only slight structural alterations. CS at 320 °C 15 min shows slightly more prominent bands of C—O groups (at 1265, 1210, and 1030 cm^{-1}) due to oxygen loss. CS obtained at 350 °C 60 min shows the flattest condition, with the almost disappearances of C—O groups. Compared to lignin, the FTIR bands were reduced in intensity rather than eliminated, indicating that the primary functional groups remain intact. The persistence but reduced intensity of C—O bands resembles trends observed in HTC carbons from saccharides [77], while lignin carbons obtained through alkali-assisted HTC often show stronger deoxygenation [19].

XRD spectra (Fig. 3c) show that the CS mainly comprise amorphous C, with some crystalline domains. Indeed, samples exhibit a predominant broad hump at $2\theta = 20°$, characteristic of amorphous carbon lacking long-range order [78]. As severity increases, particularly for the sample at 350 °C for 60 min, a more pronounced feature at $2\theta = 25°$ emerges, corresponding to (002) interlayer reflections among irregularly ordered polycyclic aromatic carbon layers within the amorphous carbon matrix [79]. The sharp peak near $2\theta = 31°$ suggests the presence of crystalline carbon or possibly other ordered carbon-containing compounds, which can be associated with graphitic carbon structures [79]. The broad hump at 20–25° is consistent with the amorphous nature of HTC carbons, whereas sharper peaks at 25° are usually observed only after activation/graphitization (>800 °C) [46]. Moreover, the CS' spectra do not exhibit the sharp peaks observed in the original lignin and attributed to inorganic species originating from lignin processing [65,80], confirming their transition from the solid to the liquid phase, thus leading to inorganic-free CS. The Raman spectra (Fig. 3d) display the typical G band at 1500–1620 cm^{-1}, attributed to the vibrational modes of sp^2 carbon atoms in defect-free graphene layers, and the D band at 1350–1400 cm^{-1}, associated with the breathing modes of rings or defects in graphitic lattices [79]. The G/D intensity ratio of ~1.8 indicates that the CS consists of predominantly amorphous and disordered carbon, with only a slight tendency toward graphitization (graphitic materials typically G/D > 10) [81]. This G/D ratio is comparable to other lignin derived CS, for example after pyrolytic stages at 700–900 °C, where G/D is 1.2–1.4 (Table 3) [82], indicating that our process yields defect-rich domains typical of low-temperature hydrothermal carbons. The variation between samples is minimal, with a slight increase in the G/D ratio observed as the temperature increases from 320 to 350 °C, suggesting minor adjustments taking place after their formation.

To the best of the authors' knowledge, the formation of spherical lignin-derived CS via hydrothermal conversion has not been reported (Table 1). Indeed, hydrothermal methods usually employ carbohydrate-rich feedstock, where 5-hydroxymethylfurfural from glucose undergoes back-polymerization to form CS, also known as secondary char [54,83]. While a solid repolymerized char is a known residue from lignin under hydrothermal liquefaction or gasification conditions [63,84], this material is rarely investigated in terms of its morphology or nanostructure. Instead, as shown in Table 3, lignin CS are typically produced via solvent-shift techniques or chemical methods, with eventual subsequent hydrothermal stages employed to stabilize the material rather than to create spherical shapes [4,18,31]. Given that the two-stage hydrothermal pathway proceeds through liquid-to-solid reactions, these CS may be regarded as a form of "lignin secondary char", similarly to that from sugars. Regarding applications, the spherical morphology, combined with the high thermal stability and richness in functional groups, can make these CS intriguing for a variety of applications, like energy storage, drug delivery or biocatalysts [18,85]. In comparison with other lignin valorization routes (Table 3), the present CS exhibit higher C content and stability than solvent-shifted particles, and finer morphological control than aerosol or chemically modified routes. While pyrolysis and activation achieve larger surface areas and higher graphitization, they require harsher conditions and corrosive reagents.

3.4. Carbon dots: photoluminescence and formation

TEM images (Fig. 2f–g and Fig. S4) reveal the presence of C-dots (size <10 nm), forming as a parallel phase to CS. Generally, C-dots are a known product of hydrothermal conversion from various biomass (Table 1) [49,93,94].

C-dots exhibit photoluminescence (PL) across excitation wavelengths 320–520 nm and have excitation-dependent emission spectra, as shown in Fig. 4 and Fig. S5. Spectra show a predominant emission is in the red region (620–700 nm), with maximum intensity occurring at excitation wavelengths of 360–400 nm. In particular, two excitation regions are evident. The first, at 320–400 nm, is characterized by shifts in PL peak positions, where shorter wavelengths likely excite heterogeneous high-energy surface states and small quantum-confined sp^2 domains. The second one at 400–520 nm, where the emission remains excitation-dependent but shows less dependence, where larger and stable sp^2 aromatic domains and subsurface regions are activated, leading to more uniform emission behavior [95]. The wavelength-

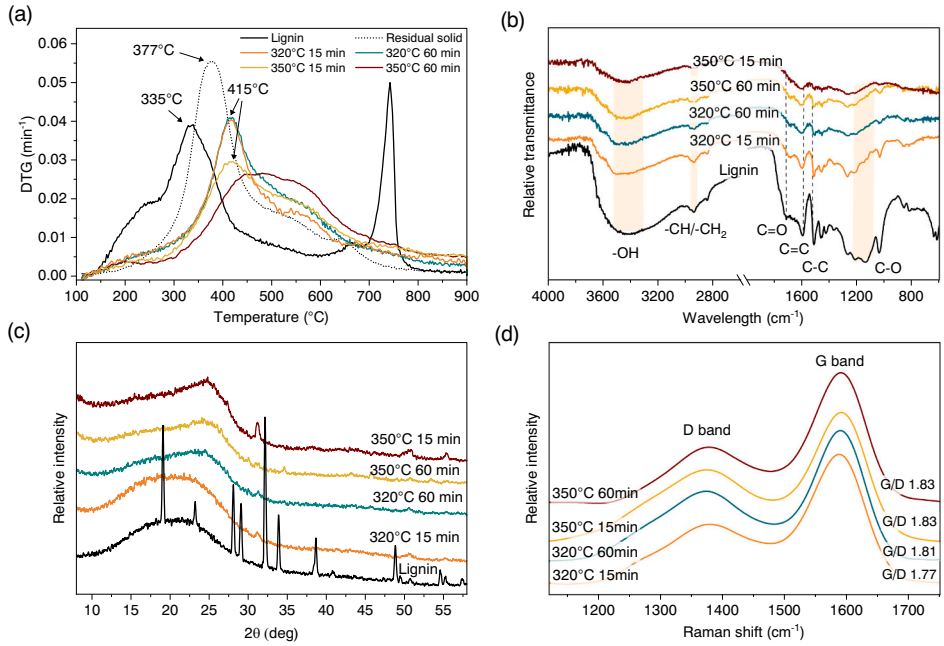

Fig. 3. Properties of CS obtained under different hydrothermal conditions: (a) DTG curves, (b) FTIR spectra, (c) XRD spectra, and (d) Raman spectra. Curves on lignin and residual solid from the first stage are also included.

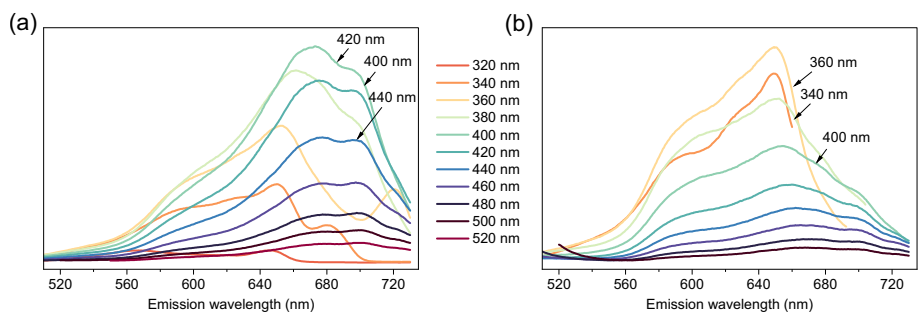

Fig. 4. PL spectra of C-dots produced at hydrothermal conditions of: (a) 320 °C 60 min; (b) 350 °C 60 min.

dependent behavior is often observed in biomass-derived C-dots and is due to surface defect states and core sp^2 structures [49,89]. A similar red-emission (at 620–700 nm) was observed in doped C-dots produced from lignin via modified hydrothermal conditions with H_2SO_4 and H_3PO_4 (Table 3) [90]. Compared to other substrates, for example, glucose-derived C-dots emit in the blue-green region (~440 nm) with sharper peaks [49].

C-dots from 320 °C exhibit a maximum PL intensity when excited at 400 nm, likely due to the activation of surface defect states or oxygen-related functional groups (e.g., C═O, C—O) [89]. Longer residence times broaden the emission range (passing from 520 to 560 nm, as time passes from 15 to 60 min). This phenomenon is likely due to the formation of larger sp^2 aromatic domains over time, along with enhanced aggregation and/or cross-linking, which contribute to the formation of diverse emissive centers [92]. The 320 °C 60 min sample has a secondary emission hump when excited <380 nm, probably due to smaller, quantum-confined sp^2 domains or high-energy surface states [49]. At 350 °C 60 min, the maximum emission shifts to 360 nm, which is likely due to an enhanced graphitization and the loss of surface defects at higher severities, leading to more uniform emissive centers with stable sp^2 domains [89]. Also, the spectrum is more uniform and free of double bands in the 340–380 nm range, suggesting a homogenization of emissive sites with severity [49]. In contrast, the 350 °C 15 min sample (Fig. S5) exhibits multiple humps and less uniformity, reflecting incomplete structural reorganization and a mix of smaller, unstable sp^2 domains and partially developed emissive centers.

C-dots likely derive from small soluble lignin fragments that undergo condensation and form polymeric intermediates, which aggregate via covalent and electrostatic bonds before carbonizing and aromatizing, as shown in previous studies [16,40]. As previously discussed, C-dots participate in the CS formation as a secondary mechanism, acting as intermediates or precursors of larger spheres via coalescence. Similarly, they also contribute to the formation of graphene-like layers (as later discussed in Section 3.5). However, it is worth emphasizing that these mechanisms are secondary and the occurrence of C-dots as transient intermediates is influenced by operating conditions - for example, longer residence times could favor their coalescence in more complex structures. While their exact solid yield can not be computed from our data, it can be estimated to be lower than that of CS, with an upper limit of 28.6 % calculated from CE data (Table S1) in the case of the absence of graphene-like layers. At longer residence times, they tend to evolve into larger carbon structures. Lower temperatures (<300 °C) and shorter residence times would likely favor their formation, as observed in other hydrothermal studies [49,93]. Compared to different routes (Table 3), lignin C-dots are usually obtained through multi-step processes with pretreatments or additives (acids, dopants, templates), which enhance yield or tune PL but reduce scalability. In contrast, the proposed two-stage hydrothermal route provides red-emissive C-dots directly from

lignin under mild, chemical free conditions, representing a greener but lower-yield alternative. Finally, it should be noted that the present reactions occur in subcritical water, where ionic pathways dominate. In contrast, radical mechanisms prevail under supercritical conditions, which have been reported to yield graphene C-dots via top-down fragmentation (Table 1) [60,96].

3.5. Graphene-like layers

TEM images (Fig. 5) reveal the presence of micron-sized graphene-like structures, with some layers partially stacked and aggregated. Images resemble the common morphology usually found in agglomerated graphene layers [4,97]. Qualitatively, TEM images show more developed graphene-like layers under more severe conditions. Some spots (Fig. 5d) show the presence of crystalline domains with interlayer spacing in the range of 1–3 nm. They are larger than those of crystalline graphite (usually ~0.34 nm [98]), indicating a significant structural disorder and crystalline domains coexisting with amorphous carbon. Analyses on the freeze-dried suspension, containing both graphene-like layers and C-dots, give indications of this product. In terms of composition (Table 4), C is 59–63 %, while O is 33–37 %, corresponding to a C/O atomic ratio of ~2.5, which falls into the range of oxide and reduced graphene oxide (rGO), which usually has a C/O of 1.9–3.1 [19,99]. FTIR spectra (Fig. S7) show prominent C═C and C—C aromatic bonds (at 1514 and 1590 cm^{-1}) and the absence of typical lignin bands, indicating the total reorganization of aromatic precursors. The minimal variation in composition across different temperatures indicates that structural changes are primarily driven by internal reorganization. XRD patterns (Fig. 6a) display a broad diffuse feature in the 2θ ~ 20–25° region and lack the sharp (002) peak of crystalline graphite or rGO, indicating disordered sp^2 carbons [73]. At 350 °C, this hump becomes slightly sharper (and tends toward ~25–26°), indicating incipient stacking/short-range ordering of aromatic layers (turbostratic (002)) rather than true graphitization [73].

Raman spectra (Fig. 6b) show broad D and G bands, indicating disordered sp^2 carbons. As shown in Table 4, the G/D ratio increases from 1.34 (at 300 °C) to ~1.8 (at 320 and 350 °C), suggesting a modest growth of sp^2 domains with temperature. All samples exhibit a weak 2D band at ~2700 cm^{-1}. 2D is generally dependent on stacking and electronic conjugation, and its presence confirms the lack of extended graphitic stacking and the prevalence of defect-rich domains instead of crystalline graphene layers, which typically exhibit a sharp Lorentzian peak [100]. Samples depart from the rGO reference (Fig. 6b), which shows a more pronounced 2D band, in agreement with the partial restoration of sp^2 conjugation upon reduction. Instead, Raman spectra more closely resemble those of graphene oxide (GO), where the 2D band appears broad and weak due to oxygen functionalities, structural disorder, and disrupted sp^2 conjugation [101–104]. In particular, the measured 2D/G intensity ratio is 0.06, indicating the presence of multilayers [101]. Indeed, the 2D band is sensitive to stacking and layer number, with the 2D/G of single-layer graphene typically >1.6 and ~0.07 for multilayers [105]. For example, Akhavan et al. measured a 2D/G of 0.82–0.89 for GO bilayer sheets [105]. Graphene like carbons via hydrothermal conversion of biomass often not even show a 2D band [19,51,53], due to extensive disorder and oxygenation. The fact that in our samples a weak but measurable feature is present indicates an equally defective yet slightly more organized structure. Other synthesis routes can yield higher values of 2D. For example, laser-assisted routes can reach a 2D/G of ~0.9–1.0 [106], carbonization followed by graphitization of >0.3 [28], and high-temperature annealing (>800–1000 °C) typically sharpens the 2D band and increases its relative intensity [107,108]. Thus, the lower ratio confirms a higher disorder and lower sp^2 domains than in other ad-hoc systems.

Therefore, in terms of formation mechanism, graphene-like layers likely form through the aromatic condensation of lignin aromatic fragments into planar and conjugated structures. Indeed, the inherent reactivity of lignin fragments, thanks to phenolic and methoxy groups, as well the high ionic product of hot-pressurized water, enables dehydration and fusion reactions. These transformations reduce steric hindrance and restore π-conjugation, driving the growth of extended aromatic domains [19,109]. It is generally known that lignin is capable of forming graphene structures via thermochemical treatments, mainly pyrolysis or solvothermal conditions (Table 3) [27,32]. For example, similar structures were also reported by Ding et al. [19,51] in an alkali-modified hydrothermal environment, while hydrothermal/solvothermal techniques are commonly used to produce graphene materials from graphitic precursors [97,109], further reinforcing our observations. Finally, the sample at 320 °C (Fig. 5a) contains clusters of C-dots organized into large micrometric foils, which disappear at 350 °C (Fig. 5b–c), leaving only graphene-like layers, suggesting that C-dots could participate in their formation. Higher temperatures probably facilitate their clustering and their reorganization into layered structures by enhancing molecular mobility and providing the thermal energy required for structural rearrangement. Overall, the similarity of the suspension with graphene-like derivatives indicates that C-dots constitute a minor fraction of the recovered material, leaving them as transient intermediates. Indeed, regarding yields, CE data (Table S1) indicate that, at 60 min, 16.8 % at 320 °C and 11.0 % at 350 °C of the C contained in lignin migrates to the liquid phase, which also contains dissolved organics. Since the freeze-dried suspension contains only graphene-like layers and C-dots (as organics evaporate during vacuum drying, thus assuming them negligible), we can provide a general estimation of their SY, equals to 26.8 at 320 °C 60 min and 17.5 % 350 °C 60 min, numbers that we can consider reasonable when dealing with production systems.

As further confirmation of the formation mechanism, we performed solvothermal runs by adding toluene at the second stage. TEM images (Fig. 5d and Fig. S6) show that its addition totally suppresses the formation of CS and C-dots, favoring the formation of stack graphene-like layers. The freeze-dried material obtained in the presence of toluene shows a higher C content (70.4–69.7 % vs. 59–63 % in hydrothermal

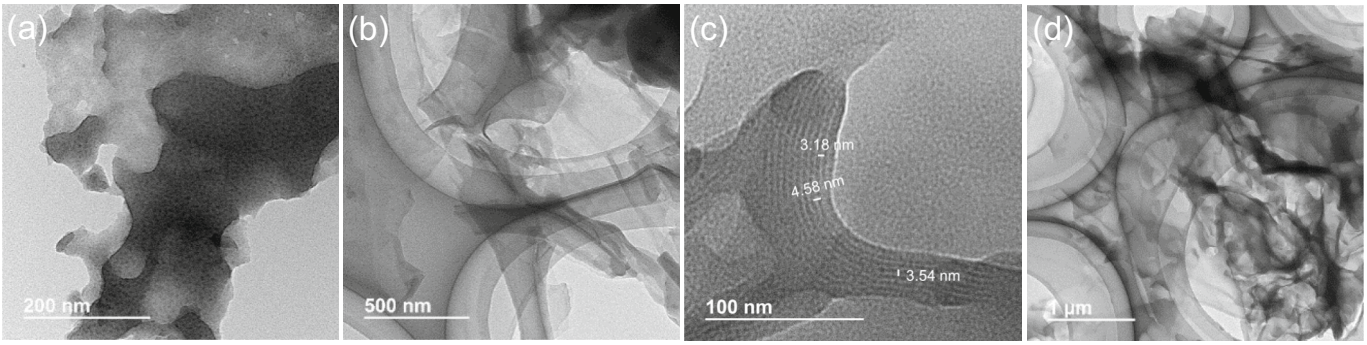

Fig. 5. TEM images of graphene-like layers produced at: (a) 320 °C, 60 min; (b–c) 350 °C, 60 min; (d) 350 °C, 60 min, with toluene addition.

Table 4
Properties of the freeze-dried suspension (containing C-dots and graphene-like layers) produced under hydrothermal and toluene-addition conditions, at a residence time of 60 min: elemental composition, H/C and O/C atomic mass ratios, intensity G/D e 2D/G ratios of Raman spectra. Confidence intervals represent standard deviations computed from error propagation (for elemental <0.5 %, H/C and O/C < 0.1). Data are referred on a mass base.

Samples		Elemental (%)				H/C (−)	O/C (−)	G/D (−)	2D/G
		C	H	N	O				(−)
Hydrothermal	300 °C	59.0	3.5	0.5	37.0	0.8	0.5	1.34	0.06
	320 °C	62.8	4.1	0.4	32.8	0.9	0.5	1.84	0.06
	350 °C	62.7	3.5	0.6	33.2	0.8	0.5	1.78	0.06
Toluene	250 °C	70.4	3.1	0.5	26.0	0.6	0.3	3.06	0.10
	320 °C	67.9	3.8	0.8	27.5	0.8	0.4	2.30	0.11
	350 °C	69.7	2.6	1.1	26.6	0.5	0.3	4.99	0.14

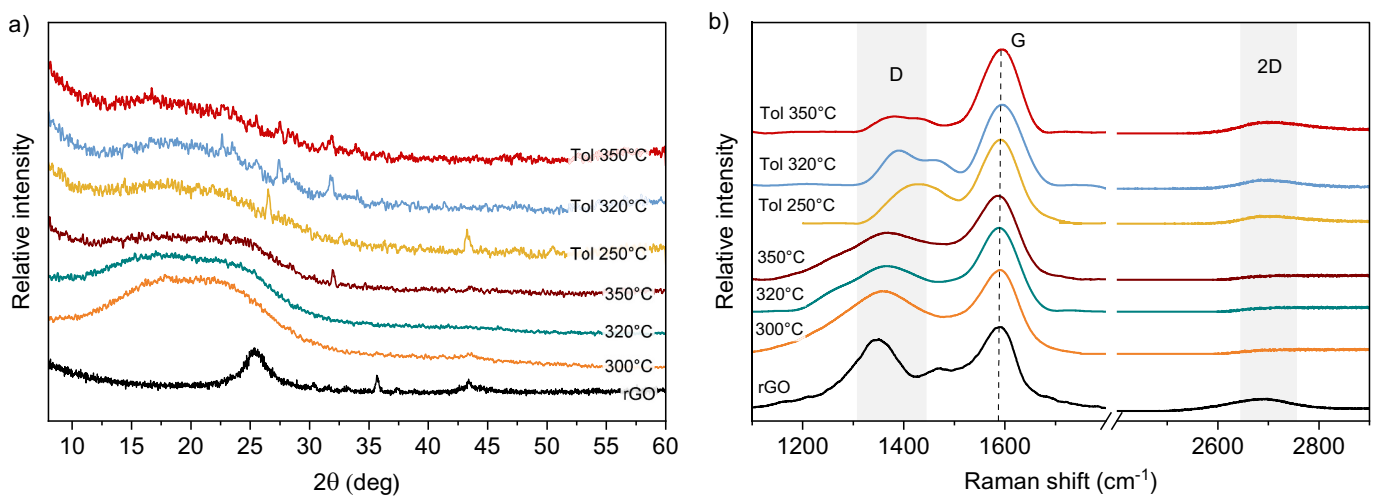

Fig. 6. Properties of the freeze-dried suspension (containing C-dots and graphene-like layers) produced at different hydrothermal temperatures at 60 min, and rGO: a) XRD spectra and b) Raman spectra.

runs, Table 4). The O/C ratio decreases to 0.3–0.4 compared to ~0.5 in hydrothermal samples, in agreement with a lower O content and partial aromatization. Although the composition does not vary strongly with temperature, higher temperatures seem to promote a more layered structure (Fig. 5), suggesting that chemical transformations occur at lower temperatures while structural reorganization and ordering take place at higher temperatures. XRD patterns (Fig. 6a) still show a broad band in the 2θ ≈ 20–25° region, typical of amorphous carbons. Although long-range graphitic order is still absent, the curve is slightly more pronounced compared to hydrothermal samples, suggesting a higher short-range stacking of aromatic layers. The 2D band of Raman spectra (Fig. 6b) remains broad and weak (centered at ~2685–2695 cm^{-1}), confirming the lack of extended crystalline domains. Nevertheless, both G/D and 2D/G ratios increase with toluene (Table 4, up to 4.99 and 0.14 at 350 °C vs. ~1.3–1.8 and ~0.06 in hydrothermal samples), consistent with reduced defect density and partial development of layered sp^2 domains. Together, these results indicate that toluene suppresses the formation of spherical intermediates and instead favors planar condensation, leading to more ordered but still defective and oxygenated graphene-like layers. This phenomenon is probably because toluene reduces hydration of intermediates, lowering the barrier for planar stacking and facilitating π–π interactions that stabilize layered growth. Also, the toluene-water mixture at temperatures below 330 °C may cause the separation of two phases, aiding in the extraction of molecules. These observations further support the proposed formation pathway: lignin aromatic fragments and C-dots act as building units for graphene-like layers, while the presence of toluene facilitates their condensation by suppressing competing routes to CS. In other words, the solvothermal environment accelerates the same structural reorganization observed in pure hydrothermal conditions, confirming that aromatic condensation and intermediate clustering are the key steps leading to layered structures.

To the best of the authors' knowledge, this is the first report of hydrothermal conversion without any additives which allowed to produce them from biomass. Indeed, common techniques (summarized in Table 3) typically involve chemical vapor deposition, plasma-based methods or the usage of templates at 1000 °C [4,53]. In the few cases where a hydrothermal treatment has been investigated (Table 1), it has been chemically modified or combined with other techniques, like in-situ hydrothermal activation using strong alkalis [51]. Then, it is worth noticing that the amorphous structure and high density of defects influence material performance. In particular, conductivity is likely low since vacancies and oxygen-containing groups interrupt sp^2 conjugation and act as scattering sites, thus lowering charge-carrier mobility [53]. The lack of long-range crystallinity (with the absence of extended graphitic sheets) weakens mechanical strength, lowering fracture resistance compared to ordered graphene sheets [27]. On the other hand, defects increase surface polarity, wettability, and the density of active sites, which is positive for applications where high conductivity is not required but surface chemistry is critical, like adsorption, catalysis, and composite reinforcement. To address specific applications (such as electrochemical or special coatings), post-treatments could be adopted. For example, annealing (>800–1000 °C), chemical reduction, or doping could decrease O content, restore sp^2 domains, and improve conductivity and mechanical properties.

3.6. Formation pathways

Fig. 7 shows an overview of the interconnected conversion of lignin to CS, C-dots, and graphene-like layers under the double-stage

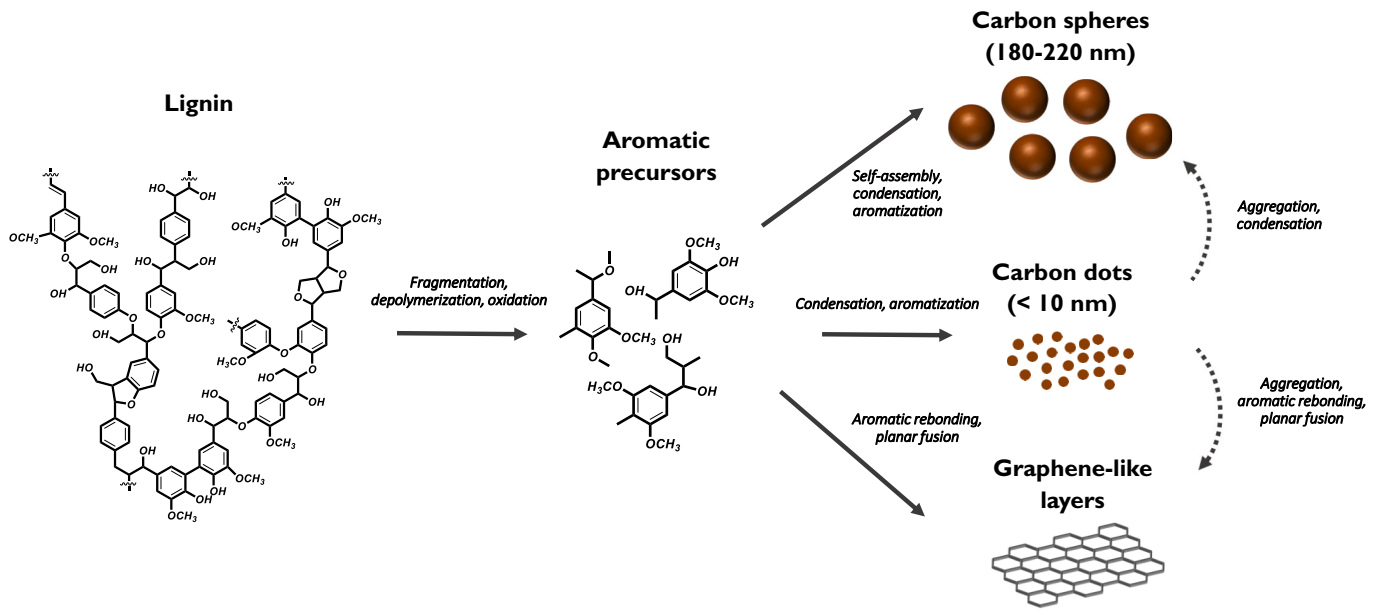

Fig. 7. Schematic overview of the formation pathways of lignin-derived carbon products via two-stage hydrothermal conversion. Solid lines indicate primary mechanisms; dashed lines indicate secondary conversion routes.

subcritical pathway. Carbon and mass balances highlight how reaction severity governs product distribution. In particular, in the first stage (250 °C, 60 min), lignin undergoes near-complete fragmentation, leaving only 5.2 % solid residue. In the second stage (300–350 °C), CS account for the major fraction (SY of 31.5–42.3 %), while graphene-like layers and C-dots contribute 26.8 % at 320 °C and 17.5 % at 350 °C (estimated from CE efficiencies, Table S1). Therefore, CS dominate at moderate severity, whereas increasing temperature shifts the balance toward layered structures and reduces the contribution of C-dots, which behave as transient intermediates. Residence time plays a secondary role: CS form already at short times (15 min) with slight variation up to 60 min, while the transition from C-dots to layered structures is mainly temperature-driven. Overall, temperature is the primary control parameter, dictating whether self-assembly into spheres, persistence of C-dots, or planar condensation into graphene-like layers prevails. Moreover, especially in the second stage, water is not only a dispersion medium but also an active reactant: its subcritical properties at 300–350 °C (high ionic product, reduced dielectric constant [39]) catalyze hydrolysis, dehydration, and condensation reactions, thereby accelerating the reorganization of lignin fragments and promoting the transition from spherical to layered carbon structures.

In more detail, the formation pathways can be summarized as follows:

- *CS* (180–220 nm): formed via self-assembly, with a secondary contribution from C-dot aggregation, followed by covalent stabilization. Self-assembly occurs already at short residence times (<15 min), driven by the amphiphilic nature of lignin: hydrophobic domains form compact cores, while hydrophilic groups stabilize the outer shell. In parallel, C-dots act as nucleation sites or intermediate structures. Then, these assemblies and aggregates, undergo further condensation and partial aromatization, resulting in thermally stable CS. The final materials exhibit SYs of 31.5–42.3 %, C of 67.8–72.5 %, and high thermal stability, making them attractive for applications such as energy storage, drug delivery, and biocatalysis [18,85].
- *C-dots* (diameters <10 nm): formed via a series of condensation reactions of small soluble lignin fragments into polymeric intermediates that aggregate via covalent and electrostatic bonds. The hydrothermal environment drives their dehydration and aromatization, stabilizing their structure. As a secondary mechanism, C-dots likely act as precursors or intermediates for larger CS and graphene-like layers. Their PL properties open potential applications in imaging, sensors, and photocatalysis [110].
- Graphene-like layers: produced from aromatic rebonding and planar fusion of lignin fragments and from C-dots. The high ionic product of water promotes dehydration and π-bonding, a process analogous to graphene oxide formation in other systems. At higher temperatures (≥350 °C), C-dots progressively disappear, leaving behind stacked, turbostratic sheets with defective but extended sp^2 domains. While highly oxygenated and disordered, these layers resemble GO-type carbons and could be exploited for conductive or catalytic applications [19,73].

3.7. Generalities and perspectives

The two-stage hydrothermal route has several advantages compared to other lignin-to-carbon materials valorization strategies (Tables 1 and 3). For example, compared to pyrolysis (≥500–900 °C) or chemical activation (≥700–900 °C with chemical agents like KOH), the process operates at lower thermal severity, uses only water as solvent, and does not require a neutralization stage for the residual liquor. CS have a nanoscale morphology and enhanced thermal stability, without any post-thermal treatment or activation. Graphene-like layers are produced without any exfoliation and high-temperature graphitization step, thereby reducing process complexity. Moreover, the absence of corrosive oxidants (such as KOH, NaOH, and H_3PO_4) and organic solvents aligns with green chemistry principles. The C-dots do not necessarily outperform those from commonly adopted hydrothermal techniques, but their spontaneous formation can make they regarded as an additional product forming in parallel with CS and graphene-like layers.

In terms of scalability, the short residence times observed here are compatible with continuous or flow hydrothermal systems, which are already established for other materials such as metal oxides [37]. The first stage, by removing most of the insoluble fraction, generates a liquid feed that is easier to pump and therefore suitable for controlled synthesis in flow reactors. In addition, as it requires only a single hydrothermal apparatus, the process is technologically simpler than multi-step routes, avoiding extra units. However, it is worth mentioning that the solid–liquid handling can represent a bottleneck, as fouling may occur during long-term continuous operation. Lignin variability is another issue, since

different technical lignins (kraft, soda, organosolv) contain various amounts of ash and structural heterogeneity, which can affect morphology and yield, requiring inlet quality control and process tuning [7,8]. Then, also batch or semi-continuous operation may remain a practical solution, especially with limited lignin stream availability. In terms of environmental impacts, the process appears feasible as a preliminary consideration. Its main advantage is the absence of chemical additives, since it relies exclusively on water as a reaction medium. Besides, using lignin as waste streams (like the black liquor from papermaking) can address waste management and circularity. In terms of energy efficiency, the two-stage hydrothermal route operates at markedly lower temperatures than those typically required for pyrolysis or activation (350 °C vs. 700–1100 °C). Although the high dilution (especially of stage 1) increases the heat load to heat up water, this drawback is partly compensated by the fact that lignin is often processed in aqueous form (e.g., black liquor [4]), which minimizes drying requirements. Moreover, pyrolysis alone is rarely employed for lignin valorization and usually needs additional activation or modification steps, thus further increasing its energy demand. Then, the energy efficiency can be improved by strategies already applied in hydrothermal systems, such as water recirculation (particularly in continuous or semi-continuous setups) or the use of renewable sources, including solar-assisted hydrothermal heating [111]. Overall, these considerations are necessarily preliminary, and future studies should focus on a more quantitative assessment of environmental impacts and energy balances to consolidate the sustainability profile of the process.

Then, several limitations should be recognized. CS and graphene-like layers show amorphous structures and are rich in defects, which can limit their electrical conductivity. Feed variability in technical lignins (kraft, soda, organosolv) can influence morphology and yield, requiring process tuning and quality control. While the materials are compositionally upgraded and thermally stable, their defect structure suggests that downstream modifications (e.g., doping or activation) may be required to enhance performance for specific applications. Nevertheless, the properties observed here already point to promising functional potential. For example, CS properties (thermal stability, morphology, and C content) are consistent with those of bio-based materials typically used in electrochemical devices (e.g., capacitors, batteries) [18]. The excitation-dependent red emission of the C-dots indicates potential in photonics, sensing, and bioimaging.

Overall, this study should be regarded as a proof of concept for the two-stage hydrothermal valorization of lignin. The findings serve as a foundation for future investigations aimed at enhancing product quality and broadening functional applications. In particular, future research could expand and improve the process in several ways:

- *Modification of the reaction environment*. We observed that the hydrophilicity and hydrophobicity of the reaction environment play a role in the equilibrium between CS and graphene-like layers. Further investigation, for example by adding solvents with different polarities, co-reactants, or catalysts, could provide some insights into their formation and optimization. It could also help in targeting specific applications such as energy storage, environmental remediation, or advanced composites.
- *Functional testing of the carbon materials*. A systematic evaluation of the electrochemical performance of CS (e.g., capacitance, conductivity, cycling stability) and the photoluminescence stability of C-dots will be essential to consolidate their applicability in specific devices. Also, adsorption capacity and surface reactivity analyses could clarify their potential in environmental remediation.
- *Kinetic monitoring at shorter timescales*. Following the early stages of lignin conversion with advanced techniques (e.g., HR-TEM, in situ spectroscopy) could provide direct insights into the transient role of C-dots and the mechanisms driving their evolution into CS and graphene-like layers.
- *Coupling with downstream modifications*. Integrating hydrothermal conversion with subsequent treatments such as thermal processing, doping, or functionalization could yield materials with enhanced properties, so addressing specific applications.
- *Continuous hydrothermal processing*. The observation that CS can form even at low residence times opens the door for exploring continuous hydrothermal processes. Precise size control through adjustments in reaction time under consistent conditions (e.g., temperature, pressure) could lead to scalable and efficient production methods for carbon materials.

4. Conclusion

This study proposes a novel method for converting lignin – a highly abundant and renewable resource – into carbon spheres, carbon dots, and graphene-like layers via a two-stage hydrothermal process. In the first stage (250 °C, 60 min), lignin undergoes defragmentation and partial depolymerization, yielding a soluble, aromatic-rich liquid phase. In the second stage (300–350 °C, 15–60 min), these aromatic precursors are converted into three distinct carbon structures. Carbon spheres (180–220 nm) form with yields of up to 42.3 %, primarily through self-assembly, and are further stabilized through condensation and aromatization. These carbon spheres exhibit high carbon content (up to 72.5 %) and enhanced thermal stability. Carbon dots (<10 nm), which confer photoluminescence to the liquid phase, are observed to coalesce and contribute to the further growth of carbon structures. In parallel, graphene-like layers – amorphous and defect-rich – form through aromatic rebonding and planar fusion, likely assisted by water's high ionic product at hydrothermal conditions.

All lignin-derived carbon products emerge within the initial 15 min of residence time, highlighting the efficiency and controllability of the process. These findings serve as a proof of concept for a fully green, scalable route to valorize lignin into functional materials. This platform opens new perspectives for circular material design from biomass residues, with potential applications across energy, catalysis, and environmental technologies.

CRediT authorship contribution statement

Giulia Ischia: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Akira Yoko:** Writing – review & editing, Validation, Project administration, Funding acquisition, Conceptualization. **Wahyudiono:** Writing – review & editing, Methodology. **Luca Fiori:** Writing – review & editing, Visualization, Supervision, Conceptualization. **Tadafumi Adschiri:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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TEM measurements with JEOL JEM-ARM200F microscope and

composition analyses of C, H, N, and O were conducted at the Analytical Research Core for Advanced Materials, Institute for Materials Research, Tohoku University. TOC analysis was conducted at Technical Division, School of Engineering, Tohoku University. This study and Giulia Ischia were supported by Invitational Fellowship Program for Collaborative Research with International Researcher for FY2023 in Tohoku University.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijbiomac.2025.147997.

Data availability

Data will be made available on request.

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